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People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
National Higher School of Mathematics



Midterm exam (1^h30)

Name : _____ Groupe: _____

Exercise 1. Among these assertions, specify which ones are true and which ones are false. (20x0.25)

	True	False
1-The median is the score at the 50th percentile.	✓	
2-The range is a frequently used measure of central tendency.		✓
3-With ordinal data, the appropriate correlation to use is the Spearman correlation.	✓	
4-The STD of a sample can be -1.		✓
5-Increasing all values of a sample by a , will increase the mean by a .	✓	
6-Increasing all values of a sample by b , will increase STD by b .		✓
7-The mean is equal to 0 for a symmetric histogram.		✓
8-If a value is higher than the mean then it is an outlier.		✓
9-The median is always higher than the mean.		✓
10-The sample mean is always higher than the population mean.		✓
11- $\rho(X, Y)$ is a unitless number between -1 and 1.	✓	
12- $\forall a; b; x_0; y_0 \in \mathbb{R} : Cov(aX + x_0; bY + y_0) < abCov(X; Y)$		✓
13-Parameter is a numerical measure that describes a variable of a population.	✓	
14-A sample is a subset of the people or objects in a population.	✓	
15-Skewed distributions usually show the familiar bell shape.		✓
16-Descriptive statistics are used to summarize and display the data in a meaningful way.	✓	
17-A correlation of 1 or -1 means that the two variables are perfectly linearly related.	✓	
18-Correlation implies causation.		✓
19-The interquartile range is the difference between the first and second quartiles.		✓
20-A scatter plot can be used to show the correlation between two continuous variables.	✓	

Exercise 2. Table below some data on the weights (in kilograms) of 35 dogs sample in a park. :

class	frequency	Rel. Frequency	Cum. Frequency	Cum. Rel. Frequency
[5, 10[	4	0.11	4	0.11
[10, 15[	6	0.17	10	0.28
[15, 20[	8	0.23	18	0.51
[20, 25[	5	0.14	23	0.66
[25, 30[	4	0.11	27	0.77
[30, 35[	3	0.09	30	0.86
[35, 40[	5	0.14	35	1

(1)

- Complete the statistical table above.
- Where is the median weight located? Give its value.

The median weight $\tilde{X} \in [15, 20[$ and given by

$$\tilde{X} = L_1 + \left(\frac{\frac{n}{2} - C f_b}{f_m} \right) \times w = 15 + \left(\frac{\frac{35}{2} - 10}{8} \right) \times (20 - 15) = \mathbf{19.688} \quad (0.75)$$

3. Give Q_1, Q_3 and interquartile range

The first quartile Q_1 is located in the class $[10, 15[$ and the third quartile Q_3 is in $[25, 30[$ and both are given by

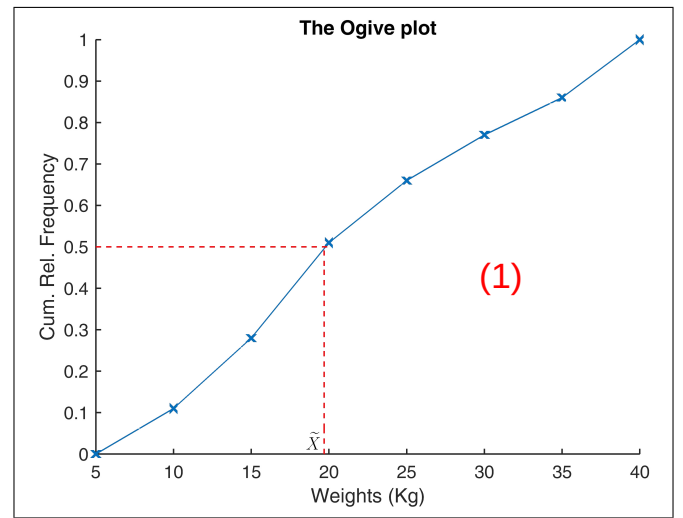
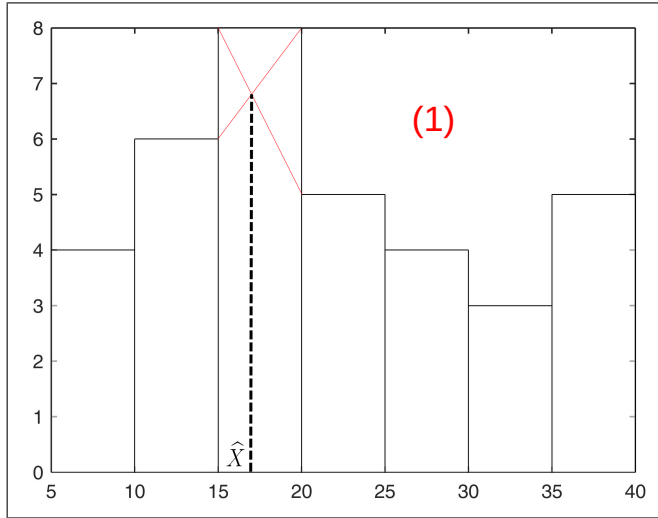
$$Q_1 = L_1 + \left(\frac{\frac{n}{4} - Cf_{q_1}}{f_{q_1}} \right) \times w = 10 + \left(\frac{\frac{35}{4} - 4}{6} \right) \times (15 - 10) = \mathbf{13.958}$$

and

$$Q_3 = L_1 + \left(\frac{\frac{3n}{4} - Cf_{q_3}}{f_{q_3}} \right) \times w = 25 + \left(\frac{\frac{105}{4} - 23}{4} \right) \times (30 - 25) = \mathbf{29.063.} \quad (1.5)$$

The interquartile range is $IQR = Q_3 - Q_1 = 29.063 - 13.958 = \mathbf{15.105}$.

4. Plot the histogram and the Ogive plot of the above distribution then determine the mode and median of the series graphically



5. Give the modale class then calculate the mode value.

Clearly, The highest frequency is for the third class $[15, 20[$ to which the mode $\hat{X}$ belongs and it's given by

$$\hat{X} = L_{mo} + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) \times w = 15 + \left(\frac{(8 - 6)}{(8 - 6) + (8 - 5)} \right) \times (20 - 15) = \mathbf{17.0} \quad (0.75)$$

6. Calculate the mean, variance, standard deviation and coefficient of variation.

Since the data are grouped, we need to compute the med point of each class and the formulas to be used are :

$$\mu = \bar{X} = \frac{\sum_{i=1}^7 f_i x_i}{\sum_{i=1}^7 f_i} = \frac{752.5}{35} = \mathbf{21.5} \quad (0.25)$$

$$\sigma^2 = Var(X) = \frac{1}{\sum_{i=1}^N f_i - 1} \sum_{i=1}^N f_i (x_i - \mu)^2 = \frac{3190}{34} = \mathbf{93.8235} \quad (0.5)$$

and

$$\sigma = \sqrt{\frac{1}{\sum_{i=1}^N f_i - 1} \sum_{i=1}^N f_i (x_i - \mu)^2} = \sqrt{93.8235} = \mathbf{9.6863.} \quad (0.25)$$

The coefficient of variation is given by

$$CV = \frac{\sigma}{\mu} \times 100 = CV = \frac{9.6863}{21.5} \times 100 = \mathbf{45.053\%} \quad (0.25)$$

7. Do you think the distribution is symmetric? justify.

No, the distribution is not symmetric, characterized by the shape of the histogram, also by computing pearson's coefficient of skewness

(0.25)

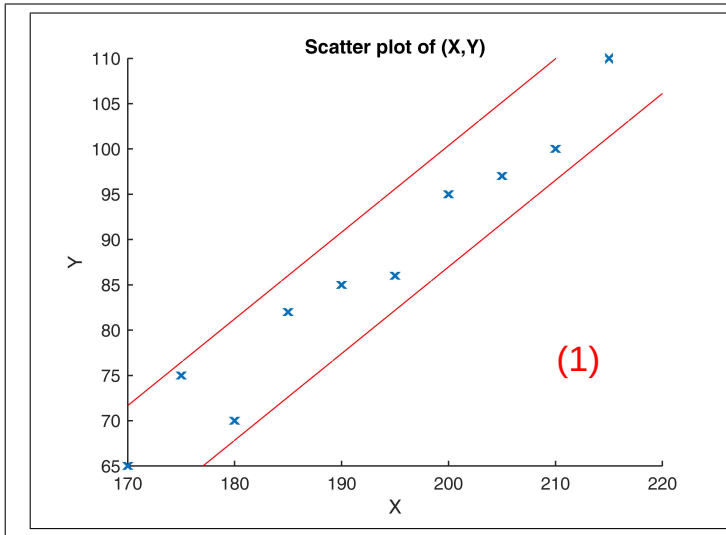
$$sk = \frac{3 \times (21.5 - 19.688)}{9.6863} = \mathbf{0.56121} \quad (0.5)$$

we can observe that its value is positive which let us conclude that the distribution is asymmetric

Exercise 3. The following results were obtained:

X	170	175	180	185	190	195	200	205	210	215
Y	65	75	70	82	85	86	95	97	100	110

1. Draw the scatter plot of the given data. Describe the relationship between X and Y.



the plotted points fall in a narrow band and show a rising tendency from the lower left hand corner to the upper right hand corner, therefore, it suggests a **High Degree of Positive Linear Correlation**

(1)

(0.5)

2. Calculate the correlation coefficient and interpret the obtained result.

The above scatter plot suggested the presence of linear correlation, hence, we choose pearson's coefficient of correlation to measure it.

We have

$$\sum x_i = 1925, \sum y_i = 865, \sum (x_i - \bar{X})^2 = 2062.5, \sum (y_i - \bar{Y})^2 = 1806.5, \sum (x_i - \bar{X})(y_i - \bar{Y}) = 1887.5.$$

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X \cdot \sigma_Y} = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum (x_i - \bar{X})^2} \sqrt{\sum (y_i - \bar{Y})^2}} = \frac{1887.5}{\sqrt{2062.5} \times \sqrt{1806.5}} = \mathbf{0.97785} \quad (2)$$

which indicates the existence of a strong positive linear correlation between the two variables. (0.5)

Exercise 4. The Spearman correlation coefficient is defined as the Pearson correlation coefficient between the rank variables

$$r_s = \rho(R(X), R(Y)) = \frac{cov(R(X), R(Y))}{\sigma_{R(X)} \sigma_{R(Y)}}.$$

For a sample of size n, if all n ranks are distinct integers, show that the above formula can be written as

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

where $d_i = R(X_i) - R(Y_i)$ is the difference between the two ranks of each observation.

Since there are no ties, the $R(X)$'s and $R(Y)$'s both consist of the integers from 1 to n inclusive. Hence we can rewrite the denominator:

$$r_s = \frac{\sum_i \left(R(X_i) - \overline{R(X)} \right) \left(R(Y_i) - \overline{R(Y)} \right)}{\sqrt{\sum_i \left(R(X_i) - \overline{R(X)} \right)^2 \sum_i \left(R(Y_i) - \overline{R(Y)} \right)^2}} = \frac{\sum_i \left(R(X_i) - \overline{R(X)} \right) \left(R(Y_i) - \overline{R(Y)} \right)}{\sum_i \left(R(X_i) - \overline{R(X)} \right)^2}$$

But the denominator is just a function of n

$$\sum_i \left(R(X_i) - \overline{R(X)} \right)^2 = \sum_i R(X_i)^2 - n \overline{R(X)}^2$$

In the special case where the $R(X_i)$ and $R(Y_i)$ are distinct ranks, each is a permutation of the same sequence of numbers $1, 2, \dots, n$. Thus $\overline{R(X)} = \overline{R(Y)} = \frac{(n+1)}{2}$ and $\sum_i R(X_i)^2 = \sum_i i^2 = \frac{n(n+1)(2n+1)}{6}$. Therefore

$$\begin{aligned} \sum_i \left(R(X_i) - \overline{R(X)} \right)^2 &= n(n+1) \left(\frac{(2n+1)}{6} - \frac{(n+1)}{4} \right) = n(n+1) \frac{(n-1)}{12} \\ &= \frac{n(n^2-1)}{12} \end{aligned}$$

Now let's look at the numerator, we have $R(X_i) R(Y_i) = \frac{1}{2} (R(X_i)^2 + R(Y_i)^2 - (R(X_i) - R(Y_i))^2)$

$$\begin{aligned} \sum_i \left(R(X_i) - \overline{R(X)} \right) \left(R(Y_i) - \overline{R(Y)} \right) &= \sum_i R(X_i) R(Y_i) - n \overline{R(X)} \overline{R(Y)} \\ &= \frac{1}{2} \sum_i (R(X_i)^2 + R(Y_i)^2 - (R(X_i) - R(Y_i))^2) - n \left(\frac{(n+1)}{2} \right)^2 \\ &= \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)^2}{4} - \frac{1}{2} \sum_i d_i^2 \\ &= \frac{n(n^2-1)}{12} - \frac{1}{2} \sum_i d_i^2. \end{aligned}$$

Finally

$$r_s = \frac{\frac{n(n^2-1)}{12} - \frac{1}{2} \sum_i d_i^2}{\frac{n(n^2-1)}{12}} = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2-1)}$$

**3pts for a complete answer only.
Partial answers are not accepted**